Deep Graph-Based Modelling of Clinical and Lifestyle Data for Diabetes Detection

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Background

Clinical setting

Diabetes represents a heterogeneous group of metabolic disorders characterized by chronic hyperglycemia, with clinically distinct subtypes such as *type 1 diabetes*, *type 2 diabetes*, *prediabetes*, and *gestational diabetes* [1]. Accurately distinguishing not only between diabetic and non-diabetic individuals but also among the specific subtypes has important diagnostic, prognostic, and therapeutic implications.

The *complex interrelationships* among patient demographics, clinical history, laboratory measurements, and treatment records can be overlooked by traditional approaches. By representing this information as hypergraphs and applying Hypergraph Neural Networks (HGNN), we can extract meaningful features to improve diabetes-specific condition prediction performance.

Hypergraphs

Usually, a *hypergraph* [2] is a generalisation of a standard graph where each hyper-edge can connect any number of nodes, enabling the modelling of higher-order relationships that cannot be captured by pairwise edges alone.

A hypergraph [3] can be formally expressed as

$$G = (V, E),$$

where $V = \{v_1, v_2, \ldots, v_{|V|}\}$ denotes the set of nodes and $E = \{e_1, e_2, \ldots, e_{|E|}\}$ the set of hyperedges. Each hyperedge e_j is defined as a non-empty subset of V, that is

$$\emptyset \neq e_j \subseteq V.$$

Alternatively, the hyperedge set can be represented by an incidence matrix

$$H \in \{0,1\}^{|V|\times|E|},$$

where the entry $H_{i,j} = 1$ if node v_i belongs to hyperedge e_j, and $H_{i,j} = 0$ otherwise.

For a given node $v_i \in V$, the set of incident hyperedges is defined as

$$N_E(v_i) = \{\, e_k \in E \mid v_i \in e_k \,\}.$$

Hypergraphs are particularly effective in modeling co-authorship networks, biological interactions, and complex patient-feature relationships in healthcare data [3].

Through HGNNs, information can be propagated across hyper-edges, enabling the capture of richer structural patterns and enhancing predictive performance. Clinical studies have leveraged HGNNs for diverse tasks, including mortality and readmission prediction [4], medical diagnosis and medication recommendation [5], as well as other healthcare applications [6].

Methods

In the proposed method, each patient was represented as a node, while hyper-edges were constructed to group patients sharing common clinical characteristics, by using clustering rules clinically based. This modelling paradigm was applied to a dataset of 100.000 patients containing multiple information including demographics, lifestyle, family and medical history, cardiovascular history, and clinical/laboratory measurements [7].

Figure 1 depicts the implemented a HGNN for diabetes prediction. The model consists of stacked hypergraph convolutional layers with Batch Normalization and Dropout and the Adam optimizer using early stopping based on validation F1-score. The task was formulated as binary classification to distinguish diabetic vs. non-diabetic patients.

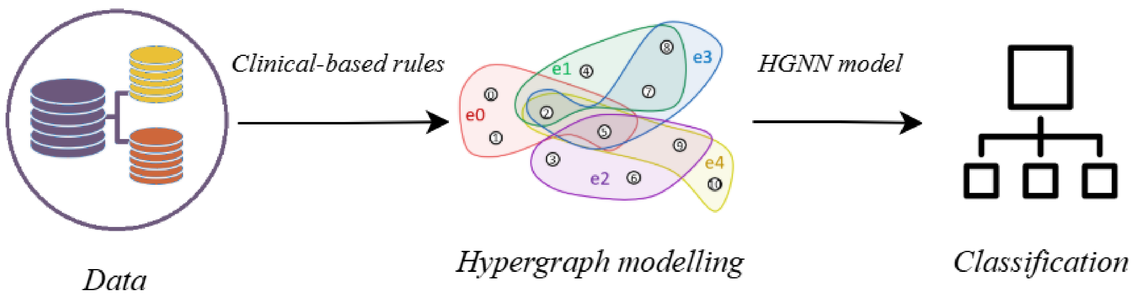

Figure 1. Proposed Workflow

Results

The model, as shown in Fig.2 and in Table 1 consistently outperformed standard fully connected baselines, indicating that incorporating multi-patient hyper-edge relationships improved the separation between diabetic and non-diabetic classes.

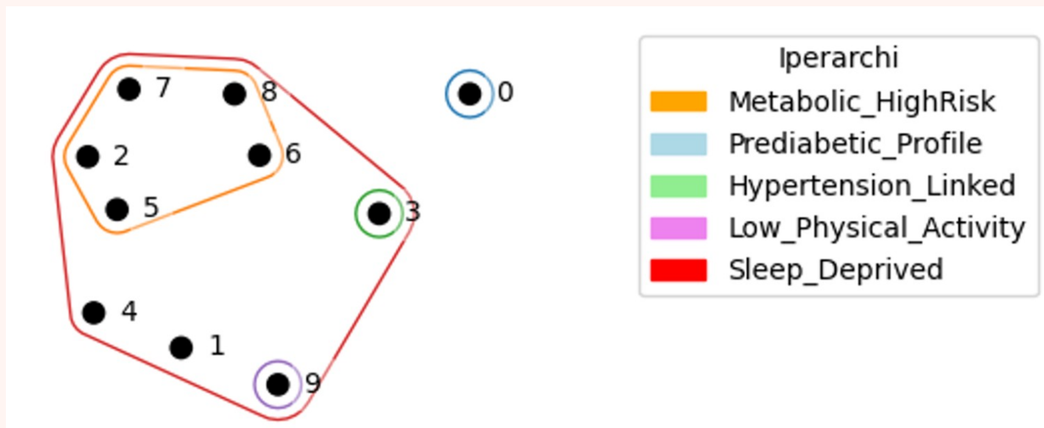

Figure 2. Obtained HyperGraph with Hypernetx occhio c e ipergrafi

Class	Precision	Recall	F1-score
Non-diabetic (0)	0.8827	0.8641	0.8733
Diabetic (1)	0.8669	0.8851	0.8759
Accuracy		**0.8746**	

Table 1. Classification performance of the HGNN model.

Conclusion

The HGNN achieved strong predictive performance in the binary setting, with high accuracy and F1-score on the test set. Future work will extend the model to multiclass classification, enabling discrimination among different diabetes subtypes rather than only diabetic vs. non-diabetic.

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